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(54) **POTATO CULTIVAR FL 2505**

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(57) **ABSTRACT**

A potato cultivar designated FL 2505 is disclosed. The invention relates to tubers of potato cultivar FL 2505, to seeds of potato cultivar FL 2505, to plants and plant parts of potato cultivar FL 2505, to food products produced from potato cultivar FL 2505, and to methods for producing a potato plant by crossing potato cultivar FL 2505 with itself or with another potato variety. The invention also relates to methods for producing a transgenic potato plant and to the transgenic potato plants and parts produced by those methods. This invention also relates to potato plants and plant parts derived from potato cultivar FL 2505, to methods for producing other potato plants or plant parts derived from potato cultivar FL 2505 and to the potato plants and their parts derived from use of those methods. The invention further relates to hybrid potato tubers, seeds, plants and plant parts produced by crossing potato cultivar FL 2505 with another potato cultivar.

19 Claims, No Drawings

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POTATO CULTIVAR FL 2505**BACKGROUND**

All publications cited in this application are herein incorporated by reference.

The embodiments recited herein relate to a novel potato cultivar designated FL 2505 and to the tubers, plants, plant parts, tissue culture and seeds produced by that potato variety. The embodiments further relate to food products produced from potato cultivar FL 2505, such as, but not limited to, french fries, potato chips, dehydrated potato material, potato flakes, and potato granules.

Potatoes are a tuberous crop grown from the perennial plant *Solanum tuberosum*. The potato is one of the top five most important food crops in the world and the leading vegetable crop in the United States (United States Department of Agriculture, Economic Research Service, updated Oct. 19, 2016).

The foregoing examples of the related art and limitations related therewith are intended to be illustrative and not exclusive. Other limitations of the related art will become apparent to those of skill in the art upon a reading of the specification.

SUMMARY

It is to be understood that the embodiments include a variety of different versions or embodiments, and this Summary is not meant to be limiting or all-inclusive. This Summary provides some general descriptions of some of the embodiments, but may also include some more specific descriptions of other embodiments.

An embodiment provides a potato cultivar designated FL 2505. Another embodiment relates to said variety being deposited with a Budapest Treaty depository, such as, Provasoli-Guillard National Center for Marine Algae and Microbiota, Bigelow Laboratory for Ocean Sciences, 60 Bigelow Drive, East Boothbay, Maine 04544, United States. Another embodiment relates to the tubers, and potato seeds of potato cultivar FL 2505, to the plants of potato cultivar FL 2505 and to methods for producing a potato plant produced by crossing potato cultivar FL 2505 with itself or another potato cultivar, and the creation of variants by mutagenesis, gene editing, or transformation of potato cultivar FL 2505.

Any such methods using potato cultivar FL 2505 are a further embodiment: selfing, backcrosses, hybrid production, crosses to populations, and the like. All plants produced using potato cultivar FL 2505 as at least one parent are within the scope of the embodiments. Advantageously, potato cultivar FL 2505 could be used in crosses with other, different potato plants to produce first generation (F₁) potato hybrid seeds and plants with superior characteristics. Another embodiment relates to a method of vegetatively propagating the tuber of FL 2505, comprising the steps of: collecting tissue or cells capable of being propagated from said plant; cultivating said tissue or cells to obtain proliferated shoots; and rooting said proliferated shoots to obtain rooted plantlets; or cultivating said tissue or cells to obtain proliferated shoots, or to obtain plantlets and a potato plant produced by growing the plantlets or proliferated shoots therefrom.

Another embodiment provides for single or multiple gene converted plants of potato cultivar FL 2505. The transferred gene(s) may be a dominant or recessive allele. The transferred gene(s) may confer such traits as herbicide resistance, insect resistance, resistance for bacterial, fungal, or viral

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disease, male fertility, male sterility, enhanced nutritional quality, modified fatty acid metabolism, modified carbohydrate metabolism, modified yield, modified glycoalkaloid content, and industrial usage. The gene may be a naturally occurring potato gene or a transgene introduced through genetic engineering techniques.

Another embodiment provides for regenerable cells for use in tissue culture of potato cultivar FL 2505. The tissue culture may be capable of regenerating plants having all the physiological and morphological characteristics of the foregoing potato plant, and of regenerating plants having substantially the same genotype as the foregoing potato plant. The regenerable cells in such tissue cultures may be embryos, protoplasts, meristematic cells, callus, pollen, leaves, ovules, anthers, cotyledons, hypocotyl, pistils, roots, root tips, flowers, seeds, tuber, light sprout, petiole, tubers, or stems. Still a further embodiment provides for potato plants regenerated from the tissue cultures of potato cultivar FL 2505.

As used herein, “at least one,” “one or more,” and “and/or” are open-ended expressions that are both conjunctive and disjunctive in operation. For example, each of the expressions “at least one of A, B and C,” “at least one of A, B, or C,” “one or more of A, B, and C,” “one or more of A, B, or C” and “A, B, and/or C” means A alone, B alone, C alone, A and B together, A and C together, B and C together, or A, B and C together.

As used herein, “sometime” means at some indefinite or indeterminate point of time. So for example, as used herein, “sometime after” means following, whether immediately following or at some indefinite or indeterminate point of time following the prior act.

Various embodiments are set forth in the Detailed Description as provided herein and as embodied by the claims. It should be understood, however, that this Summary does not contain all of the aspects and embodiments, is not meant to be limiting or restrictive in any manner, and that embodiment(s) as disclosed herein is/are understood by those of ordinary skill in the art to encompass obvious improvements and modifications thereto.

In addition to the exemplary aspects and embodiments described above, further aspects and embodiments will become apparent by study of the following descriptions.

Definitions

In the description and tables herein, a number of terms are used. In order to provide a clear and consistent understanding of the specification and claims, including the scope to be given such terms, the following definitions are provided:

Black spot. A black spot may be brown, gray, or black in appearance and is found in bruised tuber tissue as a result of a pigment called melanin that is produced following the injury of cells. Black spots occur primarily in the perimedullary tissue just beneath the vascular ring, but may be large enough to include a portion of the cortical tissue.

Cotyledon. A cotyledon is a type of seed leaf. The cotyledon contains the food storage tissues of the seed.

Embryo. The embryo is the small plant contained within a mature seed.

Golden nematode. *Globodera rosiochiensis*, commonly known as golden nematode, is a plant parasitic nematode affecting the roots and tubers of potato plants. Symptoms include poor plant growth, wilting, water stress and nutrient deficiencies.

Hypocotyl. A hypocotyl is the portion of an embryo or seedling between the cotyledons and the root. Therefore, it can be considered a transition zone between shoot and root.

Light Sprout or Sprout. Refers to the “eyes” or sprouts that grow from the buds on the surface of the potato skin.

Locus. Locus or loci (plural) refers to a position in the genome for a gene, SNP, mutation, etc.

Plant Parts. Plant parts (or a potato plant, or a part thereof) includes but is not limited to, regenerable cells in such tissue cultures may be embryos, protoplasts, meristematic cells, callus, pollen, leaves, ovules, anthers, cotyledons, hypocotyl, pistils, roots, root tips, flowers, seeds, tuber, eye, light sprout, tuber, petiole, or stems.

Progeny. Progeny includes an F₁ potato plant produced from the cross of two potato plants where at least one plant includes potato cultivar FL 2505 and progeny further includes, but is not limited to, subsequent F₂, F₃, F₄, F₅, F₆, F₇, F₈, F₉, and F₁₀ generational crosses with the recurrent parental line.

Quantitative Trait Loci (QTL). Quantitative trait loci (QTL) refer to genetic loci that control to some degree numerically representable traits that are usually continuously distributed.

Regeneration. Refers to the development of a plant from tissue culture.

RHS. RHS refers to the Royal Horticultural Society color reference.

Single Gene Converted (Conversion). Single gene converted (conversion) plants refers to plants which are developed by a plant breeding technique called backcrossing wherein essentially all of the desired morphological and physiological characteristics of a variety are recovered in addition to the single gene transferred into the variety via the backcrossing technique or via genetic engineering.

Specific gravity. Refers to an expression of density and is a measurement of potato quality. There is a high correlation between the specific gravity of the tuber and the starch content and percentage of dry matter or total solids. A higher specific gravity contributes to higher recovery rate and better quality of the processed product.

DETAILED DESCRIPTION

Potato cultivar FL 2505 originated from a private breeding program near Rhinelander, Wisconsin. FL 2505 is the result of classical hybridization breeding. In 2009, proprietary parental line FL 1867 (female parent) and FL 1953 were crossed. FL 1867 was chosen for its uniform size, high dry matter content and its potential for transmitting Golden Nematode resistance to its progeny. FL 1953 was chosen for its early maturity, bright chip color out of storage and its potential for transmitting Wart resistance to its progeny. Seeds from the cross were sown in pots near Rhinelander, Wisconsin in 2011 and the resulting tubers were harvested in the fall of 2011. The resulting progeny tubers were planted in a field in the spring of 2012, where the selection criteria were tuber set and appearance, high specific gravity and no Common Scab. A single plant was chosen and given the experimental designation ‘2012 52.02’ and subsequently named FL 2505. From 2016 to 2017, FL 2505 was planted and tested in 10 separate locations of greenhouses and fields in Rhinelander, Wisconsin and other locations in the United States, and tested for low early sugars, high yield and solids and a small profile size.

Potato cultivar FL 2505 has shown uniformity and stability, as described in the following variety description information. Potato cultivar FL 2505 was tested for uniformity via tuber propagation for six generations in Rhinelander, Wisconsin and for two generations in ten locations around the United States in randomized block replicated trials. Potato cultivar FL 2505 was tested for uniformity and stability a sufficient number of generations with careful attention to uniformity of plant type and has been increased with continued observation for uniformity. Characteristics were taken in part according to the USDA Plant Variety Protection (PVP) form.

Potato cultivar FL 2505 has the following morphologic and other characteristics based primarily on data collected in Rhinelander, Wisconsin.

TABLE 1

VARIETY DESCRIPTION INFORMATION (COMPRISED OF TABLES 1A AND 1B) TABLE 1A - CHARACTERISTICS	
Characteristic	FL 2505
Market class	Chip-processing
Light sprout, general shape	Ovoid
Light sprout base, pubescence of base	Weak
Light sprout base, anthocyanin coloration	Red-violet
Light sprout base, intensity of anthocyanin coloration	Strong
Light sprout, tip habit	Closed
Light sprout tip pubescence	Medium
Light sprout tip anthocyanin coloration	Red-violet
Light sprout tip, intensity of anthocyanin coloration	Weak
Light sprout root initials, frequency	Some
Plant growth habit	Semi-erect
Plant type	Stem (foliage open and stems clearly visible)
Plant maturity (days after planting at vine senescence)	140
Maturity class	Mid-season (111-120 days after planting)
Regional area	North Central
Stem anthocyanin coloration	Weak
Stem wings	Medium
Leaf color (fully developed leaf located on middle 1/3 of plant)	Medium-green, RHS N137B

TABLE 1-continued

VARIETY DESCRIPTION INFORMATION (COMPRISED OF TABLES 1A AND 1B) TABLE 1A - CHARACTERISTICS	
Characteristic	FL 2505
Leaf silhouette	Open
Petiole, anthocyanin coloration	Medium
Leaf stipule size	Medium
Terminal leaflet shape	Medium-ovate
Terminal leaflet apex shape	Acuminate
Terminal leaflet base shape	Truncate
Terminal leaflet margin waviness	Weak
Average number of primary leaflet pairs	4.325 (range is 4 to 5)
Primary leaflet apex shape	Acuminate
Primary leaflet size	Large
Primary leaflet shape	Medium-ovate
Primary leaflet base shape	Cordate
Average number of secondary and tertiary leaflet pairs	11.05 (range is 8 to 17)
Average number of inflorescences per plant	3.925 (range is 1 to 6)
Average number of florets per inflorescence	5.3 (range is 1 to 12)
Corolla color	Inner surface is RHS N80A (Purple-violet) and outer surface is RHS N80C (Purple- violet)
Corolla shape	Pentagonal
Calyx anthocyanin coloration	Weak
Anther color	RHS 14A
Anther shape	Pear-shaped cone
Stigma shape	Clavate
Stigma color	RHS 137
Tuber, predominant skin color	RHS 199B (Tan)
Tuber, secondary skin color	Absent
Tuber skin texture	Rough (flaky)
Tuber shape	Round
Average tuber thickness	Medium-thick; 60.2 mm
Average tuber length	60.5 mm (range is 29.7 mm to 84.6 mm)
Average weight of individual tuber	5.36 g
Average tuber width	59.7 mm (range is 24.4 mm to 96 mm)
Tuber eye depth	Shallow
Tuber lateral eyes	Shallow
Distribution of tuber eyes	Predominantly apical
Prominence of tuber eyebrows	Slight prominence
Predominant tuber flesh color	RHS 155B (White)
Secondary tuber flesh color	Absent
Number of tubers per plant	Medium; 8 to 15
Total glycoalkaloid content	11.3 mg/100 g fresh tuber
Specific gravity	Greater than 1.090

TABLE 1B

PEST AND DISEASE RESPONSES	
Disease	FL 2505
Late blight (<i>Phytophthora</i>)	Susceptible
Early blight (<i>Alternaria</i>)	Moderately susceptible
Soft rot (<i>Erwinia</i>)	Susceptible
Common scab (<i>Streptomyces</i>)	Intermediate susceptible
Powdery scab (<i>Spongospora</i>)	Susceptible
Dry rot (<i>Fusarium</i>)	Intermediate susceptible
Potato Virus Y (PVY)	Susceptible
Golden Nematode (<i>Globodera</i>)	Susceptible

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Table 2 shows differences between Potato Cultivar FL 2505 and potato cultivar FL 1867.

TABLE 2

DIFFERENCES BETWEEN FL 2505 AND FL 1867		
Characteristic	FL 2505	FL 1867
Flower color	Purple-violet	White
Vine maturity	Late (140 days after planting)	Mid (119 days after planting)

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Breeding With Potato Cultivar FL 2505

The complexity of inheritance influences choice of the breeding method. Backcross breeding is used to transfer one or a few favorable genes for a highly heritable trait into a desirable cultivar. This approach has been used extensively for breeding disease-resistant cultivars. Various recurrent selection techniques are used to improve quantitatively inherited traits controlled by numerous genes. The use of recurrent selection in self-pollinating crops depends on the ease of pollination, the frequency of successful hybrids from each pollination, and the number of hybrid offspring from each successful cross.

Promising advanced breeding lines are thoroughly tested and compared to appropriate standards in environments representative of the commercial target area(s) for three or more years. The best lines are candidates for new commercial cultivars; those still deficient in a few traits may be used as parents to produce new populations for further selection.

These processes, which lead to the final step of marketing and distribution, usually take from eight to twelve years from the time the first cross is made. Therefore, development of new cultivars is a time-consuming process that requires precise forward planning, efficient use of resources, and a minimum of changes in direction.

A most difficult task is the identification of individuals that are genetically superior, because for most traits the true genotypic value is masked by other confounding plant traits or environmental factors. One method of identifying a superior plant is to observe its performance relative to other experimental plants and to a widely grown standard cultivar. If a single observation is inconclusive, replicated observations provide a better estimate of its genetic worth.

The goal of potato breeding is to develop new and superior potato cultivars and hybrids. The breeder initially selects and crosses two or more parental lines, followed by repeated selfing and selection, producing many new genetic combinations. The breeder can theoretically generate billions of different genetic combinations via crossing, selection, selfing and mutations.

The development of new potato cultivars requires the development and selection of potato varieties, the crossing of these varieties and selection of superior hybrid crosses. The hybrid seed is produced by manual crosses between selected male-fertile parents or by using male sterility systems. These hybrids are selected for certain single gene traits such as pod color, flower color, pubescence color or herbicide resistance which indicate that the seed is truly a hybrid. Additional data on parental lines, as well as the phenotype of the hybrid, influence the breeder's decision whether to continue with the specific hybrid cross.

Breeding programs combine desirable traits from two or more cultivars or various broad-based sources into breeding pools from which cultivars are developed by selfing and selection of desired phenotypes. Pedigree breeding is used commonly for the improvement of self-pollinating crops. Two parents that possess favorable, complementary traits are crossed to produce an F_1 . An F_2 population is produced by selfing one or several F_1 plants. Selection of the best individuals may begin in the F_2 population; then, beginning in the F_3 , the best individuals in the best families are selected. Replicated testing of families can begin in the F_4 generation to improve the effectiveness of selection for traits with low heritability. At an advanced stage of inbreeding (i.e., F_6 and F_7), the best lines or mixtures of phenotypically similar lines are tested for potential release as new cultivars. Using Potato Cultivar FL 2505 to Develop Other Potato Varieties

Potato varieties such as potato cultivar FL 2505 are typically developed for use in seed and tuber production. However, potato varieties such as potato cultivar FL 2505 also provide a source of breeding material that may be used to develop new potato varieties. Plant breeding techniques known in the art and used in a potato breeding program include, but are not limited to, recurrent selection, mass selection, bulk selection, mass selection, backcrossing, pedigree breeding, open pollination breeding, restriction fragment length polymorphism enhanced selection, genetic marker enhanced selection, making double haploids, transformation, and gene editing. These techniques can be used singularly or in combinations. The development of potato varieties in a breeding program requires, in general, the development and evaluation of homozygous varieties. There are many analytical methods available to evaluate a new variety. The oldest and most traditional method of analysis is the observation of phenotypic traits, but genotypic analysis may also be used.

Additional Breeding Methods

One embodiment is directed to methods for producing a potato plant by crossing a first parent potato plant with a second parent potato plant, wherein the first or second potato plant is the potato plant from potato cultivar FL 2505. Further, both first and second parent potato plants may be from potato cultivar FL 2505. Any plants produced using potato cultivar FL 2505 as at least one parent are also within the scope of the embodiments. These methods are well known in the art and some of the more commonly used breeding methods are described herein. Descriptions of breeding methods can be found in one of several reference books (e.g., Allard, *Principles of Plant Breeding* (1960); Simmonds, *Principles of Crop Improvement* (1979); Snee, et al. (1979); Cooper, S. G., D. S. Douches and E. J. Grafius. 2004. Combining genetic engineering and traditional breeding to provide elevated resistance in potatoes to Colorado potato beetle. *Entom. Exper. Applic.* 112:37-46; Ross, H. 1986. *Potato Breeding—Problems and Perspectives*. Advances in Plant Breeding. Suppl. 13. J. Plant Breed. Verlag. Paul Parey, Berlin).

The following describes breeding methods that may be used with potato cultivar FL 2505 in the development of further potato plants. One such embodiment is a method for developing a potato cultivar FL 2505 progeny plant in a potato breeding program comprising: obtaining the potato plant, or a part thereof, of potato cultivar FL 2505, utilizing said plant, or plant part, as a source of breeding material, and selecting a potato cultivar FL 2505 progeny plant with molecular markers in common with potato cultivar FL 2505 and/or with morphological and/or physiological characteristics selected from the characteristics listed in Tables 1 and/or 2. Breeding steps that may be used in the potato plant breeding program include pedigree breeding, backcrossing, mutation breeding, and recurrent selection. In conjunction with these steps, techniques such as RFLP-enhanced selection, genetic marker enhanced selection (for example, SSR markers), and the making of double haploids may be utilized.

Another method involves producing a population of potato cultivar FL 2505 progeny potato plants, comprising crossing potato cultivar FL 2505 with another potato plant, thereby producing a population of potato plants which derive 50% of their alleles from potato cultivar FL 2505. A plant of this population may be selected and repeatedly selfed or sibbed with a potato cultivar resulting from these successive filial generations. One embodiment is the potato cultivar produced by this method and that has obtained at

least 50% of its alleles from potato cultivar FL 2505. See, Milbourne, D., et al. "Comparison of PCR-based marker systems for the analysis of genetic relationships in cultivated potato" in *Molecular Breeding*, 3(2): 127-136 (April 1997); Jacobs, J. M. E, et al., "genetic map of potato (*Solanum tuberosum*) integrating molecular markers, including transposons, and classical markers" *Theoretical and Applied Genetics*, 91(2): 289-300 (July 1995).

One of ordinary skill in the art of plant breeding would know how to evaluate the traits of two plant varieties to determine if there is no significant difference between the two traits expressed by those varieties. For example, see, Fehr and Walt, *Principles of Cultivar Development*, pp. 261-286 (1987). Thus, embodiments include potato cultivar FL 2505 progeny potato plants comprising a combination of at least two potato cultivar FL 2505 traits selected from the group consisting of those listed in Tables 1 and 2 and a combination of traits listed in the Summary, so that said progeny potato plant is not significantly different for said traits than potato cultivar FL 2505 as determined at the 5% significance level when grown in the same environmental conditions. Using techniques described herein, molecular markers may be used to identify said progeny plant as a potato cultivar FL 2505 progeny plant. Mean trait values may be used to determine whether trait differences are significant, and the traits are measured on plants grown under the same environmental conditions. Once such a variety is developed, its value is substantial since it is important to advance the germplasm base as a whole in order to maintain or improve traits such as yield, disease resistance, pest resistance, and plant performance in extreme environmental conditions.

Progeny of potato cultivar FL 2505 may also be characterized through their filial relationship with potato cultivar FL 2505, as for example, being within a certain number of breeding crosses of potato cultivar FL 2505. A breeding cross is a cross made to introduce new genetics into the progeny, and is distinguished from a self or a sib cross, which is made to select among existing genetic alleles. The lower the number of breeding crosses in the pedigree, the closer the relationship between potato cultivar FL 2505 and its progeny. For example, progeny produced by the methods described herein may be within 1, 2, 3, 4, or 5 breeding crosses of potato cultivar FL 2505.

Pedigree Breeding

Pedigree breeding starts with the crossing of two genotypes, such as potato cultivar FL 2505 and another potato variety having one or more desirable characteristics that is lacking or which complements potato cultivar FL 2505. If the two original parents do not provide all the desired characteristics, other sources can be included in the breeding population. In the pedigree method, superior plants are selfed and selected in successive filial generations. In the succeeding filial generations, the heterozygous condition gives way to homogeneous varieties as a result of self-pollination and selection. Typically, in the pedigree method of breeding, five or more successive filial generations of selfing and selection is practiced: F_1 to F_2 ; F_2 to F_3 ; F_3 to F_4 ; F_4 to F_5 ; etc. After a sufficient amount of inbreeding, successive filial generations will serve to increase seed of the developed variety.

Backcross Breeding

Backcross breeding has been used to transfer genes for a simply inherited, highly heritable trait into a desirable homozygous cultivar or inbred line which is the recurrent parent. The source of the trait to be transferred is called the donor parent. After the initial cross, individuals possessing

the phenotype of the donor parent are selected and repeatedly crossed (backcrossed) to the recurrent parent. The resulting plant is expected to have the attributes of the recurrent parent (e.g., cultivar) and the desirable trait transferred from the donor parent. This is also known as single gene conversion.

The selection of a suitable recurrent parent is an important step for a successful backcrossing procedure. The goal of a backcross protocol is to alter or substitute a single trait or characteristic in the original variety. To accomplish this, a single gene of the recurrent variety is modified or substituted with the desired gene from the nonrecurrent parent, while retaining essentially all of the rest of the desired genetic, and therefore the desired physiological and morphological constitution of the original variety. The choice of the particular nonrecurrent parent will depend on the purpose of the backcross; one of the major purposes is to add some agronomically important trait to the plant. The exact backcrossing protocol will depend on the characteristic or trait being altered to determine an appropriate testing protocol. Although backcrossing methods are simplified when the characteristic being transferred is a dominant allele, a recessive allele may also be transferred. In this instance, it may be necessary to introduce a test of the progeny to determine if the desired characteristic has been successfully transferred.

Many single gene traits have been identified that are not regularly selected for in the development of a new variety but that can be improved by backcrossing techniques well-known in the art. Single gene traits may or may not be transgenic. Examples of these traits include, but are not limited to, herbicide resistance, insect resistance, resistance for bacterial, fungal, or viral disease, male fertility, male sterility, enhanced nutritional quality, modified fatty acid metabolism, modified carbohydrate metabolism, modified yield, modified glycoalkaloid content, and industrial usage.

In addition to being used to create a backcross conversion, backcrossing can also be used in combination with pedigree breeding. As discussed previously, backcrossing can be used to transfer one or more specifically desirable traits from one variety, the donor parent, to a developed variety called the recurrent parent, which has overall good agronomic characteristics yet lacks that desirable trait or traits. However, the same procedure can be used to move the progeny toward the genotype of the recurrent parent, but at the same time retain many components of the nonrecurrent parent by stopping the backcrossing at an early stage and proceeding with selfing and selection. For example, a potato variety may be crossed with another variety to produce a first generation progeny plant. The first generation progeny plant may then be backcrossed to one of its parent varieties to create a BC_1 or BC_2 . Progeny are selfed and selected so that the newly developed variety has many of the attributes of the recurrent parent and yet several of the desired attributes of the nonrecurrent parent. This approach leverages the value and strengths of the recurrent parent for use in new potato varieties.

Therefore, an embodiment of the present disclosure is a method of making a backcross conversion potato cultivar FL 2505, comprising the steps of crossing a plant of potato cultivar FL 2505 with a donor plant comprising a desired trait, selecting an F_1 progeny plant comprising the desired trait, and backcrossing the selected F_1 progeny plant to a plant of potato cultivar FL 2505 to produce BC_1 , BC_2 , BC_3 , etc. This method may further comprise the step of obtaining a molecular marker profile of potato cultivar FL 2505 and using the molecular marker profile to select for a progeny

plant with the desired trait and the molecular marker profile of potato cultivar FL 2505. In one embodiment, the desired trait is a mutant gene, gene, or transgene present in the donor parent.

Recurrent Selection and Mass Selection

Recurrent selection is a method used in a plant breeding program to improve a population of plants. Potato cultivar FL 2505 is suitable for use in a recurrent selection program. The method entails individual plants cross pollinating with each other to form progeny. The progeny are grown and the superior progeny selected by any number of selection methods, which include individual plant, half-sib progeny, full-sib progeny, and selfed progeny. The selected progeny are cross pollinated with each other to form progeny for another population. This population is planted and again superior plants are selected to cross pollinate with each other. Recurrent selection is a cyclical process and therefore can be repeated as many times as desired. The objective of recurrent selection is to improve the traits of a population. The improved population can then be used as a source of breeding material to obtain new varieties for commercial or breeding use, including the production of a synthetic cultivar. A synthetic cultivar is the resultant progeny formed by the intercrossing of several selected varieties.

Mass selection is a useful technique when used in conjunction with molecular marker enhanced selection. In mass selection, seeds from individuals are selected based on phenotype or genotype. These selected seeds are then bulked and used to grow the next generation. Bulk selection requires growing a population of plants in a bulk plot, allowing the plants to self-pollinate, harvesting the seed in bulk, and then using a sample of the seed harvested in bulk to plant the next generation. Also, instead of self-pollination, directed pollination could be used as part of the breeding program.

Mass and recurrent selections can be used to improve populations of either self- or cross-pollinating crops. A genetically variable population of heterozygous individuals is either identified, or created, by intercrossing several different parents. The plants are selected based on individual superiority, outstanding progeny, or excellent combining ability. The selected plants are intercrossed to produce a new population in which further cycles of selection are continued.

Single-Seed Descent

The single-seed descent procedure in the strict sense refers to planting a segregating population, harvesting a sample of one seed per plant, and using the one-seed sample to plant the next generation. When the population has been advanced from the F_2 to the desired level of inbreeding, the plants from which lines are derived will each trace to different F_2 individuals. The number of plants in a population declines each generation due to failure of some seeds to germinate or some plants to produce at least one seed. As a result, not all of the F_2 plants originally sampled in the population will be represented by a progeny when generation advance is completed.

Mutation Breeding

Mutation breeding is another method of introducing new traits into potato cultivar FL 2505. Mutations that occur spontaneously or are artificially induced can be useful sources of variability for a plant breeder. The goal of artificial mutagenesis is to increase the rate of mutation for a desired characteristic. Mutation rates can be increased by many different means including temperature, long-term seed storage, tissue culture conditions, radiation; such as X-rays, Gamma rays (e.g., cobalt 60 or cesium 137), neutrons,

(product of nuclear fission by uranium 235 in an atomic reactor), Beta radiation (emitted from radioisotopes such as phosphorus 32 or carbon 14), or ultraviolet radiation (preferably from 2500 to 2900 nm), or chemical mutagens (such as base analogues (5-bromo-uracil)), related compounds (8-ethoxy caffeine), antibiotics (streptonigrin), alkylating agents (sulfur mustards, nitrogen mustards, epoxides, ethylenamines, sulfates, sulfonates, sulfones, lactones), azide, hydroxylamine, nitrous acid, or acridines. Once a desired trait is observed through mutagenesis the trait may then be incorporated into existing germplasm by traditional breeding techniques. Details of mutation breeding can be found in Fehr, "Principles of Cultivar Development," Macmillan Publishing Company (1993). In addition, mutations created in other potato plants may be used to produce a backcross conversion of potato cultivar FL 2505 that comprises such mutation.

Additional methods include, but are not limited to, expression vectors introduced into plant tissues using a direct gene transfer method, such as microprojectile-mediated delivery, DNA injection, electroporation, and the like. Expression vectors may be introduced into plant tissues by using either microprojectile-mediated delivery with a biolistic device or by using *Agrobacterium*-mediated transformation. Transformant plants obtained with the protoplasm of the embodiments are intended to be within the scope of the embodiments.

Gene Editing Using CRISPR

Targeted gene editing can be done using CRISPR/Cas9 technology (Saunders & Joung, *Nature Biotechnology*, 32, 347-355, 2014). CRISPR is a type of genome editing system that stands for Clustered Regularly Interspaced Short Palindromic Repeats. This system and CRISPR-associated (Cas) genes enable organisms, such as select bacteria and archaea, to respond to and eliminate invading genetic material. Ishino, Y., et al. *J. Bacteriol.* 169, 5429-5433 (1987). These repeats were known as early as the 1980s in *E. coli*, but Barrangou and colleagues demonstrated that *S. thermophilus* can acquire resistance against a bacteriophage by integrating a fragment of a genome of an infectious virus into its CRISPR locus. Barrangou, R., et al. *Science* 315, 1709-1712 (2007). Many plants have already been modified using the CRISPR system. See for example, U.S. Application Publication No. WO2014068346 (György et al., Identification of a *Xanthomonas euvesicatoria* resistance gene from pepper (*Capsicum annuum*) and method for generating plants with resistance); Martinelli, F. et al., "Proposal of a Genome Editing System for Genetic Resistance to Tomato Spotted Wilt Virus" *American Journal of Applied Sciences* 2014; Noman, A. et al., "CRISPR-Cas9: Tool for Qualitative and Quantitative Plant Genome Editing" *Frontiers in Plant Science* Vol. 7 Nov. 2016; and "Exploiting the CRISPR/Cas9 System for Targeted Genome Mutagenesis in *Petunia*" *Science Reports* Volume 6: February 2016; Potatoes have been modified using the CRISPR system. Please see Wang, S., et al., "Efficient targeted mutagenesis in potato by the CRISPR/Cas9 system" *Plant Cell Reports* 34(9): pp 1473-1476 (September 2015).

Gene editing can also be done using crRNA-guided surveillance systems for gene editing. Additional information about crRNA-guided surveillance complex systems for gene editing can be found in the following documents, which are incorporated by reference in their entirety: U.S. Application Publication No. 2010/0076057 (Sontheimer et al., Target DNA Interference with crRNA); U.S. Application Publication No. 2014/0179006 (Feng, CRISPR-CAS Component Systems, Methods, and Compositions for Sequence

Manipulation); U.S. Application Publication No. 2014/0294773 (Brouns et al., Modified Cascade Ribonucleoproteins and Uses Thereof); Sorek et al., *Annu. Rev. Biochem.* 82:273-266, 2013; and Wang, S. et al., *Plant Cell Rep* (2015) 34: 1473-1476. Therefore, it is another embodiment to use the CRISPR system on potato variety FL 2505 to modify traits and resistances or tolerances to pests, herbicides, diseases, and viruses.

Gene Editing Using TALENs

Transcription activator-like effector nucleases (TALENs) have been successfully used to introduce targeted mutations via repair of double stranded breaks (DSBs) either through non-homologous end joining (NHEJ), or by homology-directed repair (HDR) and homology-independent repair in the presence of a donor template. Thus, TALENs are another mechanism for targeted genome editing using MZ-H002. The technique is well known in the art; see for example Malzahn, Aimee et al. "Plant genome editing with TALEN and CRISPR" *Cell & bioscience* vol. 7 21. 24 Apr. 2017.

Therefore, it is another embodiment to use the TALENs system on potato variety FL 2505 to modify traits and resistances or tolerances to pests, herbicides, and viruses.

Other Methods of Genome Editing

In addition to CRISPR and TALENs, two other types of engineered nucleases can be used for genome editing: engineered homing endonucleases/meganucleases (EMNs), and zinc finger nucleases (ZFNs). These methods are well known in the art. See for example, Petilino, Joseph F. "Genome editing in plants via designed zinc finger nucleases" *In Vitro Cell Dev Biol Plant.* 51(1): pp. 1-8 (2015); and Daboussi, Fayza, et al. "Engineering Meganuclease for Precise Plant Genome Modification" in *Advances in New Technology for Targeted Modification of Plant Genomes*. Springer Science+Business. pp 21-38 (2015).

Therefore, it is another embodiment to use engineered nucleases on potato variety FL 2505 to modify traits and resistances or tolerances to pests, herbicides, and viruses. Introduction of a New Trait or Locus into Potato Cultivar FL 2505

Potato cultivar FL 2505 represents a new variety into which a new locus or trait may be introgressed. Direct transformation and backcrossing represent two important methods that can be used to accomplish such an introgression. The term backcross conversion and single locus conversion are used interchangeably to designate the product of a backcrossing program.

Backcross Conversions of Potato Cultivar FL 2505

A backcross conversion of potato cultivar FL 2505 occurs when DNA sequences are introduced through backcrossing (The Potato Genome Sequencing Consortium, "Genome sequence and analysis of the tuber crop potato" *Nature*. 475: 189-195. (14 Jul. 2011); Hallauer, et al., "Corn Breeding," *Corn and Corn Improvements*, No. 18, pp. 463-481 (1988)), with potato cultivar FL 2505 utilized as the recurrent parent. Both naturally occurring and transgenic DNA sequences may be introduced through backcrossing techniques. A backcross conversion may produce a plant with a trait or locus conversion in at least two or more backcrosses, including at least 2 crosses, at least 3 crosses, at least 4 crosses, at least 5 crosses, and the like. Molecular marker assisted breeding or selection may be utilized to reduce the number of backcrosses necessary to achieve the backcross conversion. For example, see, Barone, Amalia, "Molecular marker-assisted selection for potato breeding" *American Journal of Potato Research*. 81(2):111-117 (March 2004), and Openshaw, S. J., et al., Marker-assisted Selection in Backcross Breeding, Proceedings Symposium of the Analy-

sis of Molecular Data, Crop Science Society of America, Corvallis, Oreg. (August 1994), where it is demonstrated that a backcross conversion can be made in as few as two backcrosses.

The complexity of the backcross conversion method depends on the type of trait being transferred (single genes or closely linked genes as compared to unlinked genes), the level of expression of the trait, the type of inheritance (cytoplasmic or nuclear), and the types of parents included in the cross. It is understood by those of ordinary skill in the art that for single gene traits that are relatively easy to classify, the backcross method is effective and relatively easy to manage. Desired traits that may be transferred through backcross conversion include, but are not limited to, sterility (nuclear and cytoplasmic), fertility restoration, nutritional enhancements, drought tolerance, nitrogen utilization, altered fatty acid profile, low phytate, industrial enhancements, disease resistance (bacterial, fungal, or viral), insect resistance, and herbicide resistance. In addition, an introgression site itself, such as an FRT site, Lox site, or other site specific integration site, may be inserted by backcrossing and utilized for direct insertion of one or more genes of interest into a specific plant variety. In some embodiments, the number of loci that may be backcrossed into potato cultivar FL 2505 is at least 1, 2, 3, 4, or 5, and/or no more than 6, 5, 4, 3, or 2. A single locus may contain several transgenes, such as a transgene for disease resistance that, in the same expression vector, also contains a transgene for herbicide resistance. The gene for herbicide resistance may be used as a selectable marker and/or as a phenotypic trait. A single locus conversion of site specific integration system allows for the integration of multiple genes at the converted loci.

The backcross conversion may result from either the transfer of a dominant allele or a recessive allele. Selection of progeny containing the trait of interest is accomplished by direct selection for a trait associated with a dominant allele. Transgenes transferred via backcrossing typically function as a dominant single gene trait and are relatively easy to classify. Selection of progeny for a trait that is transferred via a recessive allele requires growing and selfing the first backcross generation to determine which plants carry the recessive alleles. Recessive traits may require additional progeny testing in successive backcross generations to determine the presence of the locus of interest. The last backcross generation is usually selfed to give pure breeding progeny for the gene(s) being transferred, although a backcross conversion with a stably introgressed trait may also be maintained by further backcrossing to the recurrent parent with selection for the converted trait.

Along with selection for the trait of interest, progeny are selected for the phenotype of the recurrent parent. The backcross is a form of inbreeding, and the features of the recurrent parent are automatically recovered after successive backcrosses. Poehlman, *Breeding Field Crops*, p. 204 (1987). Poehlman suggests from one to four or more backcrosses, but as noted above, the number of backcrosses necessary can be reduced with the use of molecular markers. Other factors, such as a genetically similar donor parent, may also reduce the number of backcrosses necessary. As noted by Poehlman, backcrossing is easiest for simply inherited, dominant, and easily recognized traits.

One process for adding or modifying a trait or locus in potato cultivar FL 2505 comprises crossing potato cultivar FL 2505 plants grown from potato cultivar FL 2505 seed with plants of another potato variety that comprise the desired trait or locus, selecting F₁ progeny plants that

comprise the desired trait or locus to produce selected F₁ progeny plants, crossing the selected progeny plants with the potato cultivar FL 2505 plants to produce backcross progeny plants, selecting for backcross progeny plants that have the desired trait or locus and the morphological characteristics of potato cultivar FL 2505 to produce selected backcross progeny plants, and backcrossing to potato cultivar FL 2505 three or more times in succession to produce selected fourth or higher backcross progeny plants that comprise said trait or locus. The modified potato cultivar FL 2505 may be further characterized as having the physiological and morphological characteristics of potato cultivar FL 2505 listed in Tables 1 and 2 and the Summary as determined at the 5% significance level when grown in the same environmental conditions and/or may be characterized by percent similarity or identity to potato cultivar FL 2505 as determined by SSR markers. The above method may be utilized with fewer backcrosses in appropriate situations, such as when the donor parent is highly related or markers are used in the selection step. Desired traits that may be used include those nucleic acids known in the art, some of which are listed herein, that will affect traits through nucleic acid expression or inhibition. Desired loci include the introgression of FRT, Lox, and other sites for site specific integration, which may also affect a desired trait if a functional nucleic acid is inserted at the integration site.

In addition, the above process and other similar processes described herein may be used to produce first generation progeny potato seed by adding a step at the end of the process that comprises crossing potato cultivar FL 2505 with the introgressed trait or locus with a different potato plant and harvesting the resultant first-generation progeny potato seed.

Molecular Techniques Using Potato Cultivar FL 2505

The advent of new molecular biological techniques has allowed the isolation and characterization of genetic elements with specific functions, such as encoding specific protein products. Scientists in the field of plant biology developed a strong interest in engineering the genome of plants to contain and express foreign genetic elements, or additional, or modified versions of native or endogenous genetic elements in order to "alter" (the utilization of up-regulation, down-regulation, or gene silencing) the traits of a plant in a specific manner. Any DNA sequences, whether from a different species or from the same species, which are introduced into the genome using transformation or various breeding methods are referred to herein collectively as "transgenes." In some embodiments, a transgenic variant of potato cultivar FL 2505 may contain at least one transgene. Over the last fifteen to twenty years several methods for producing transgenic plants have been developed, and another embodiment also relates to transgenic variants of the claimed potato cultivar FL 2505.

Nucleic acids or polynucleotides refer to RNA or DNA that is linear or branched, single or double stranded, or a hybrid thereof. The term also encompasses RNA/DNA hybrids. These terms also encompass untranslated sequence located at both the 3' and 5' ends of the coding region of the gene: at least about 1000 nucleotides of sequence upstream from the 5' end of the coding region and at least about 200 nucleotides of sequence downstream from the 3' end of the coding region of the gene. Less common bases, such as inosine, 5-methylcytosine, 6-methyladenine, hypoxanthine and others can also be used for antisense, dsRNA and ribozyme pairing. For example, polynucleotides that contain C-5 propyne analogues of uridine and cytidine have been shown to bind RNA with high affinity and to be potent

antisense inhibitors of gene expression. Other modifications, such as modification to the phosphodiester backbone, or the 2'-hydroxy in the ribose sugar group of the RNA can also be made. The antisense polynucleotides and ribozymes can consist entirely of ribonucleotides, or can contain mixed ribonucleotides and deoxyribonucleotides. The polynucleotides of the embodiments may be produced by any means, including genomic preparations, cDNA preparations, in-vitro synthesis, RT-PCR, and in vitro or in vivo transcription.

One embodiment is a process for producing potato cultivar FL 2505 further comprising a desired trait, said process comprising introducing a transgene that confers a desired trait to a potato plant of potato cultivar FL 2505. Another embodiment is the product produced by this process. In one embodiment, the desired trait may be one or more of herbicide resistance, insect resistance, disease resistance, decreased phytate, or modified fatty acid or carbohydrate metabolism. The specific gene may be any known in the art or listed herein, including: a polynucleotide conferring resistance to imidazolinone, dicamba, sulfonyleurea, glyphosate, glufosinate, triazine, PPO-inhibitor herbicides, benzonitrile, cyclohexanedione, phenoxy propionic acid, and L-phosphinothricin; a polynucleotide encoding a *Bacillus thuringiensis* polypeptide; a polynucleotide encoding phytase, FAD-2, FAD-3, galactinol synthase, or a raffinose synthetic enzyme; or a polynucleotide conferring resistance to *Phytophthora* late blight, *Alternaria* early blight, *Erwinia* soft rot, *Streptomyces* common scab, *Spongiospora* powdery scab, *Fusarium* dry rot, Potato Leaf Roll Virus (PLRV), *Globodera rostochiensis*, or *Globodera pallida*.

Numerous methods for plant transformation have been developed, including biological and physical plant transformation protocols. See, for example, Miki et al., "Procedures for Introducing Foreign DNA into Plants," in Methods in Plant Molecular Biology and Biotechnology, Glick and Thompson Eds., CRC Press, Inc., Boca Raton, pp. 67-88 (1993), and Armstrong, "The First Decade of Maize Transformation: A Review and Future Perspective," *Maydica*, 44:101-109 (1999). In addition, expression vectors and in vitro culture methods for plant cell or tissue transformation and regeneration of plants are available. See, for example, Gruber, et al., "Vectors for Plant Transformation," in Methods in Plant Molecular Biology and Biotechnology, Glick and Thompson Eds., CRC Press, Inc., Boca Raton, pp. 89-119 (1993).

A genetic trait which has been engineered into the genome of a particular potato plant may then be moved into the genome of another variety using traditional breeding techniques that are well known in the plant breeding arts. For example, a backcrossing approach is commonly used to move a transgene from a transformed potato variety into an already developed potato variety, and the resulting backcross conversion plant would then comprise the transgene (s).

Various genetic elements can be introduced into the plant genome using transformation. These elements include, but are not limited to, genes, coding sequences, inducible, constitutive and tissue specific promoters, enhancing sequences, and signal and targeting sequences. For example, see the traits, genes, and transformation methods listed in U.S. Pat. No. 6,118,055.

Breeding with Molecular Markers

Molecular markers, which includes markers identified through the use of techniques such as Isozyme Electrophoresis, Restriction Fragment Length Polymorphisms (RFLPs), Randomly Amplified Polymorphic DNAs (RAPDs), Arbitrarily Primed Polymerase Chain Reaction

(AP-PCR), DNA Amplification Fingerprinting (DAF), Sequence Characterized Amplified Regions (SCARs), Amplified Fragment Length Polymorphisms (AFLPs), Simple Sequence Repeats (SSRs), and Single Nucleotide Polymorphisms (SNPs) may be used in plant breeding methods utilizing potato cultivar FL 2505.

Isozyme Electrophoresis and RFLPs have been widely used to determine genetic composition. See Kennedy, L. S., et al., "Identification of Sweet Potato Cultivars Using Isozyme Analysis" *HortScience* 26(3):300-302. (1991).

SSR technology can be routinely used. See Gebhardt, C., et al., "RFLP Map of the Potato" in R. L. Philipps and I. K. Vasil (eds.), *DNA-Based Markers in Plants*, 319-336, Kluwer Academic Publishers (2001).

Single Nucleotide Polymorphisms (SNPs) may also be used to identify the unique genetic composition of the embodiment(s) and progeny varieties retaining that unique genetic composition. See Vos, Peter G., et al., "Development and analysis of a 20K SNP array for potato (*Solanum tuberosum*): an insight into the breeding history" *Theor. Appl. Genet.* 128(12):2387-2401 (2015).

One use of molecular markers is Quantitative Trait Loci (QTL) mapping. QTL mapping is the use of markers, which are known to be closely linked to alleles that have measurable effects on a quantitative trait. Selection in the breeding process is based upon the accumulation of markers linked to the positive effecting alleles and/or the elimination of the markers linked to the negative effecting alleles from the plant's genome. See Danan, S., et al., "Construction of a potato consensus map and QTL meta-analysis offer new insights into the genetic architecture of late blight resistance and plant maturity traits" *BMC Plant Biol.* 2011 Jan. 19; 11:16; and Manrique-Carpintero, N. C., et al., "Genetic Map and QTL Analysis of Agronomic Traits in a Diploid Potato Population using Single Nucleotide Polymorphism Markers Molecular" *Crop Sci.* 55:2566-2579 (2015). QTL markers can also be used during the breeding process for the selection of qualitative traits. For example, markers closely linked to alleles or markers containing sequences within the actual alleles of interest can be used to select plants that contain the alleles of interest during a backcrossing breeding program. The markers can also be used to select for the genome of the recurrent parent and against the genome of the donor parent. See, Milbourne, D., et al., "Comparison of PCR-based marker systems for the analysis of genetic relationships in cultivated potato" in *Molecular Breeding*. 3(2): 127-136 (April 1997); Jacobs, J. M. E., et al., "genetic map of potato (*Solanum tuberosum*) integrating molecular markers, including transposons, and classical markers" *Theoretical and Applied Genetics*. 91(2): 289-300 (July 1995). Using this procedure can minimize the amount of genome from the donor parent that remains in the selected plants. It can also be used to reduce the number of crosses back to the recurrent parent needed in a backcrossing program. The use of molecular markers in the selection process is often called genetic marker enhanced selection. Molecular markers may also be used to identify and exclude certain sources of germplasm as parental varieties or ancestors of a plant by providing a means of tracking genetic profiles through crosses.

Production of Double Haploids

The production of double haploids can also be used for the development of plants with a homozygous phenotype in the breeding program. For example, a potato plant for which potato cultivar FL 2505 is a parent can be used to produce double haploid plants. Double haploids are produced by the doubling of a set of chromosomes (1N) from a heterozygous

plant to produce a completely homozygous individual. For example, see, Rokka, V. N. "Potato haploids in Breeding" in A. Touraev et al. (eds.) *Advances in Haploid Production in Higher Plants*, Springer Science+Business Media B.V. (2009), Chapter 17; and De Maine, M. J. "Potato Haploid Technologies" in M. Maluszynski et al. (eds), *Doubled Haploid Production in Crop Plants*, pp 241-247 (2003). This can be advantageous because the process omits the generations of selfing needed to obtain a homozygous plant from a heterozygous source.

Thus, an embodiment is a process for making a substantially homozygous potato cultivar FL 2505 progeny plant by producing or obtaining a seed from the cross of potato cultivar FL 2505 and another potato plant and applying double haploid methods to the F₁ seed or F₁ plant or to any successive filial generation.

In particular, a process of making seed retaining the molecular marker profile of potato variety FL 2505 is contemplated, such process comprising obtaining or producing F₁ seed for which potato variety FL 2505 is a parent, inducing doubled haploids to create progeny without the occurrence of meiotic segregation, obtaining the molecular marker profile of potato variety FL 2505, and selecting progeny that retain the molecular marker profile of potato variety FL 2505.

Expression Vectors for Potato Transformation: Marker Genes

Plant transformation involves the construction of an expression vector which will function in plant cells. Such a vector comprises DNA comprising a gene under control of, or operatively linked to, a regulatory element (for example, a promoter). Expression vectors include at least one genetic marker operably linked to a regulatory element (for example, a promoter) that allows transformed cells containing the marker to be either recovered by negative selection, i.e., inhibiting growth of cells that do not contain the selectable marker gene, or by positive selection, i.e., screening for the product encoded by the genetic marker. Many commonly used selectable marker genes for plant transformation are well-known in the transformation arts, and include, for example, genes that code for enzymes that metabolically detoxify a selective chemical agent which may be an antibiotic or an herbicide, or genes that encode an altered target which is insensitive to the inhibitor. A few positive selection methods are also known in the art.

One commonly used selectable marker gene for plant transformation is the neomycin phosphotransferase II (nptII) gene which, when under the control of plant regulatory signals, confers resistance to kanamycin. Lecardonnel, Anne, et al., "Genetic transformation of potato with nptII-gus marker genes enhances foliage consumption by Colorado potato beetle larvae" in *Molecular Breeding* October 1999, Volume 5, Issue 5, pp 441-451.

Another commonly used selectable marker gene is the hygromycin phosphotransferase gene which confers resistance to the antibiotic hygromycin. Kim, Hyun-Soon, et al. "The UDP-N-acetylglucosamine:Dolichol Phosphate-N-acetylglucosamine-phosphotransferase Gene as a New Selection Marker for Potato Transformation" *Biosci. Biotechnol. Biochem.*, 77(7), 1589-1592 (2013).

Additional selectable marker genes include Pain1-9a and Pain1-8c which both correspond to the group a alleles of the vacuolar acid invertase gene; Pain1prom-d e; Stp23-8b, StpL-3b, and StpL-3e which originate from two plastid starch phosphorylase genes; AGPsS-9a which is positively associated an increase in tuber starch content, starch yield and chip quality, and AGPsS-10a which is associated with a

decrease in the average tuber starch content, starch yield and chip quality; GP171-a which corresponds to allele 1a of ribulose biphosphate carboxylase activase; and Rca-1a. See Li, Li, et al., "Validation of candidate gene markers for marker-assisted selection of potato cultivars with improved tuber quality" *Theor Appl Genet.* 2013 April; 126(4): 1039-1052.

Selectable marker genes for plant transformation not of bacterial origin include, for example, mouse dihydrofolate reductase, plant 5-enolpyruvylshikimate-3-phosphate synthase, and plant acetolactate synthase (Eichholtz, et al., *Somatic Cell Mol. Genet.*, 13:67 (1987); Shah, et al., *Science*, 233:478 (1986); Charest, et al., *Plant Cell Rep.*, 8:643 (1990)).

Another class of marker genes for plant transformation requires screening of presumptively transformed plant cells, rather than direct genetic selection of transformed cells, for resistance to a toxic substance such as an antibiotic. These genes are particularly useful to quantify or visualize the spatial pattern of expression of a gene in specific tissues and are frequently referred to as reporter genes because they can be fused to a gene or gene regulatory sequence for the investigation of gene expression. Commonly used marker genes for screening presumptively transformed cells include β -glucuronidase (GUS), β -galactosidase, luciferase, and chloramphenicol acetyltransferase (Jefferson, R. A., *Plant Mol. Biol. Rep.*, 5:387 (1987); Teeri, et al., *EMBO J.*, 8:343 (1989); Koncz, et al., *Proc. Natl. Acad. Sci. USA*, 84:131 (1987); DeBlock, et al., *EMBO J.*, 3:1681 (1984)).

Expression Vectors for Potato Transformation: Promoters

Genes included in expression vectors must be driven by a nucleotide sequence comprising a regulatory element (for example, a promoter). Several types of promoters are well known in the transformation arts as are other regulatory elements that can be used alone or in combination with promoters.

As used herein, "promoter" includes reference to a region of DNA upstream from the start of transcription and involved in recognition and binding of RNA polymerase and other proteins to initiate transcription. A "plant promoter" is a promoter capable of initiating transcription in plant cells. Examples of promoters under developmental control include promoters that preferentially initiate transcription in certain tissues, such as leaves, roots, seeds, fibers, xylem vessels, tracheids, or sclerenchyma. Such promoters are referred to as "tissue-preferred." Promoters that initiate transcription only in a certain tissue are referred to as "tissue-specific." A "cell-type" specific promoter primarily drives expression in certain cell types in one or more organs, for example, vascular cells in roots or leaves. An "inducible" promoter is a promoter which is under environmental control. Examples of environmental conditions that may affect transcription by inducible promoters include anaerobic conditions or the presence of light. Tissue-specific, tissue-preferred, cell-type specific, and inducible promoters constitute the class of "non-constitutive" promoters. A "constitutive" promoter is a promoter that is active under most environmental conditions.

A. Inducible Promoters: An inducible promoter is operably linked to a gene for expression in potatoes. Optionally, the inducible promoter is operably linked to a nucleotide sequence encoding a signal sequence which is operably linked to a gene for expression in potatoes. With an inducible promoter, the rate of transcription increases in response to an inducing agent.

Any inducible promoter can be used in one or more embodiments. See, Ward, et al., *Plant Mol. Biol.*, 22:361-

366 (1993). Exemplary inducible promoters include, but are not limited to a stress-inducible *Arabidopsis* rd29A promoter, Pino, M. T., et al., "Use of a stress inducible promoter to drive ectopic AtCBF expression improves potato freezing tolerance while minimizing negative effects on tuber yield" *Plant Biotechnol. J.* 2007 September; 5(5):591-604; a light-inducible promoter Lhca3, Meiyalaghan, S., et al., "Expression of cry1 Ac9 and cry9Aa2 genes under a potato light-inducible Lhca3 promoter in transgenic potatoes for tuber moth resistance" *Euphytica* 147(3) (April 2006).

B. Constitutive Promoters: A constitutive promoter is operably linked to a gene for expression in potatoes or the constitutive promoter is operably linked to a nucleotide sequence encoding a signal sequence which is operably linked to a gene for expression in potatoes.

Many different constitutive promoters can be utilized in one or more embodiments. Exemplary constitutive promoters include, but are not limited to, the promoters from plant viruses such as the 35S promoter from CaMV (Odell, et al., *Nature*, 313:810-812 (1985)) and the promoters from such genes as rice actin (McElroy, et al., *Plant Cell*, 2:163-171 (1990)); ubiquitin (Christensen, et al., *Plant Mol. Biol.*, 12:619-632 (1989); Christensen, et al., *Plant Mol. Biol.*, 18:675-689 (1992)); pEMU (Last, et al., *Theor. Appl. Genet.*, 81:581-588 (1991)); MAS (Velten, et al., *EMBO J.*, 3:2723-2730 (1984)); and maize H3 histone (Lepetit, et al., *Mol. Gen. Genetics*, 231:276-285 (1992); Atanassova, et al., *Plant Journal*, 2 (3):291-300 (1992)). The ALS promoter, XbaI/NcoI fragment 5' to the *Brassica napus* ALS3 structural gene (or a nucleotide sequence similarity to said XbaI/NcoI fragment), represents another useful constitutive promoter. See also, U.S. Pat. No. 5,659,026.

C. Tissue-Specific or Tissue-Preferred Promoters: A tissue-specific promoter is operably linked to a gene for expression in potato. Optionally, the tissue-specific promoter is operably linked to a nucleotide sequence encoding a signal sequence which is operably linked to a gene for expression in potato. Plants transformed with a gene of interest operably linked to a tissue-specific promoter produce the protein product of the transgene exclusively, or preferentially, in a specific tissue.

Any tissue-specific or tissue-preferred promoter can be utilized in an embodiment(s). Exemplary tissue-specific or tissue-preferred promoters include, but are not limited to the C(4)-PEP(promoter, see Ghasimi, H., et al., "Green-tissue-specific, C(4)-PEPC-promoter-driven expression of Cry1Ab makes transgenic potato plants resistant to tuber moth (*Phthorimaea operculella*, Zeller), *Plant Cell Rep.* 2009 Dec. 28, (12): 1869-79; and see Lim, C. J., et al., "Screening of Tissue-Specific Genes and Promoters in Tomato by Comparing Genome Wide Expression Profiles of *Arabidopsis* Orthologues", *Mol Cells*. 2012 Jul. 31; 34(1): 53-59. Signal Sequences for Targeting Proteins to Subcellular Compartments

Transport of a protein produced by transgenes to a subcellular compartment, such as the chloroplast, vacuole, peroxisome, glyoxysome, cell wall, or mitochondrion, or for secretion into the apoplast, is accomplished by means of operably linking the nucleotide sequence encoding a signal sequence to the 5' and/or 3' region of a gene encoding the protein of interest. Targeting sequences at the 5' and/or 3' end of the structural gene may determine during protein synthesis and processing where the encoded protein is ultimately compartmentalized.

The presence of a signal sequence directs a polypeptide to either an intracellular organelle or subcellular compartment or for secretion to the apoplast. Many signal sequences are

well-known in the art. See, for example, Becker, et al., *Plant Mol. Biol.*, 20:49 (1992); Knox, C., et al., *Plant Mol. Biol.*, 9:3-17 (1987); Lerner, et al., *Plant Physiol.*, 91:124-129 (1989); Frontes, et al., *Plant Cell*, 3:483-496 (1991); Mat-suoka, et al., *Proc. Natl. Acad. Sci.*, 88:834 (1991); Gould, et al., *J. Cell. Biol.*, 108: 1657 (1989); Creissen, et al., *Plant J.*, 2:129 (1991); Kalderon, et al., *Cell*, 39:499-509 (1984); Steifel, et al., *Plant Cell*, 2:785-793 (1990).

Foreign Protein Genes and Agronomic Genes: Transformation

With transgenic plants, according to one embodiment, a foreign protein can be produced in commercial quantities. Thus, techniques for the selection and propagation of transformed plants, which are well understood in the art, yield a plurality of transgenic plants which are harvested in a conventional manner, and a foreign protein can then be extracted from a tissue of interest or from total biomass. Protein extraction from plant biomass can be accomplished by known methods which are discussed, for example, by Heney and Orr, *Anal. Biochem.*, 114:92-6 (1981).

According to an embodiment, the transgenic plant provided for commercial production of foreign protein is a potato plant. In another embodiment, the biomass of interest is potato tubers and potato seed. For the relatively small number of transgenic plants that show higher levels of expression, a genetic map can be generated, primarily via conventional RFLP, PCR, and SSR analysis, which identifies the approximate chromosomal location of the integrated DNA molecule. For exemplary methodologies in this regard, see, Glick and Thompson, *Methods in Plant Molecular Biology and Biotechnology*, CRC Press, Inc., Boca Raton, 269:284 (1993). Map information concerning chromosomal location is useful for proprietary protection of a subject transgenic plant.

Likewise, by means of one embodiment, plants can be genetically engineered to express various phenotypes of agronomic interest. Through the transformation of potato, the expression of genes can be altered to enhance disease resistance, insect resistance, herbicide resistance, agronomic, tuber quality, and other traits. Transformation can also be used to insert DNA sequences which control or help control male-sterility. DNA sequences native to potatoes, as well as non-native DNA sequences, can be transformed into potatoes and used to alter levels of native or non-native proteins. Various promoters, targeting sequences, enhancing sequences, and other DNA sequences can be inserted into the genome for the purpose of altering the expression of proteins. The interruption or suppression of the expression of a gene at the level of transcription or translation (also known as gene silencing or gene suppression) is desirable for several aspects of genetic engineering in plants.

Many techniques for gene silencing are well-known to one of skill in the art, including, but not limited to, knock-outs (such as by insertion of a transposable element such as Mu (Vicki Chandler, *The Maize Handbook*, Ch. 118 (Springer-Verlag 1994)) or other genetic elements such as a FRT, Lox, or other site specific integration sites; antisense technology (see, e.g., Sheehy, et al., *PNAS USA*, 85:8805-8809 (1988) and U.S. Pat. Nos. 5,107,065, 5,453,566, and 5,759,829); co-suppression (e.g., Taylor, *Plant Cell*, 9:1245 (1997); Jorgensen, *Trends Biotech.*, 8(12):340-344 (1990); Flavell, *PNAS USA*, 91:3490-3496 (1994); Finnegan, et al., *Bio Technology*, 12:883-888 (1994); Neuhuber, et al., *Mol. Gen. Genet.*, 244:230-241 (1994)); RNA interference (Napoli, et al., *Plant Cell*, 2:279-289 (1990); U.S. Pat. No. 5,034,323; Sharp, *Genes Dev.*, 13:139-141 (1999); Zamore, et al., *Cell*, 101:25-33 (2000); Montgomery, et al., *PNAS*

USA, 95:15502-15507 (1998)), virus-induced gene silencing (Burton, et al., *Plant Cell*, 12:691-705 (2000); Baulcombe, *Curr. Op. Plant Bio.*, 2:109-113 (1999)); target-RNA-specific ribozymes (Haseloff, et al., *Nature*, 334:585-591 (1988)); hairpin structures (Smith, et al., *Nature*, 407:319-320 (2000); U.S. Pat. Nos. 6,423,885, 7,138,565, 6,753,139, and 7,713,715); MicroRNA (Aukerman & Sakai, *Plant Cell*, 15:2730-2741 (2003)); ribozymes (Steinecke, et al., *EMBO J.*, 11:1525 (1992); Perriman, et al., *Antisense Res. Dev.*, 3:253 (1993)); oligonucleotide mediated targeted modification (e.g., U.S. Pat. Nos. 6,528,700 and 6,911,575); Zn-finger targeted molecules (e.g., U.S. Pat. Nos. 7,151,201, 6,453,242, 6,785,613, 7,177,766 and 7,788,044); and other methods or combinations of the above methods known to those of skill in the art.

Additional Methods for Potato Transformation

Numerous methods for plant transformation have been developed including biological and physical plant transformation protocols. See, for example, Chakaravarty, B., et al., "Genetic transformation in potato: Approaches and strategies" *American Journal of Potato Research* 84(4):301-311.

A. *Agrobacterium*-mediated Transformation: One method for introducing an expression vector into plants is based on the natural transformation system of *Agrobacterium*. See, for example, Orozco-Cárdenas, M. L., et al., (2014). Potato (*Solanum tuberosum* L.) Methods in Molecular biology, *Agrobacterium* Protocols edited by Kan Wang. Third Edition Volume 2, Humana Press. Totowa, New Jersey, (2014).

B. Direct Gene Transfer: Several methods of plant transformation, collectively referred to as direct gene transfer, have been developed as an alternative to *Agrobacterium*-mediated transformation. A generally applicable method of plant transformation is microprojectile-mediated transformation where DNA is carried on the surface of microprojectiles measuring 1 to 4 μ m. The expression vector is introduced into plant tissues with a biolistic device that accelerates the microprojectiles to speeds of 300 to 600 m/s which is sufficient to penetrate plant cell walls and membranes. Sanford, et al., *Part. Sci. Technol.*, 5:27 (1987); Sanford, J. C., *Trends Biotech.*, 6:299 (1988); Klein, et al., *Bio Tech.*, 6:559-563 (1988); Sanford, J. C., *Physiol Plant*, 7:206 (1990); Klein, et al., *Biotechnology*, 10:268 (1992). See also, U.S. Pat. No. 5,015,580 (Christou, et al.), issued May 14, 1991 and U.S. Pat. No. 5,322,783.

Another method for physical delivery of DNA to plants is sonication of target cells. Zhang, et al., *Bio Technology*, 9:996 (1991). Alternatively, liposome and spheroplast fusion have been used to introduce expression vectors into plants. Deshayes, et al., *EMBO J.*, 4:2731 (1985); Christou, et al., *Proc Natl. Acad. Sci. USA*, 84:3962 (1987). Direct uptake of DNA into protoplasts using CaCl_2 precipitation, polyvinyl alcohol or poly-L-ornithine have also been reported. Hain, et al., *Mol. Gen. Genet.*, 199:161 (1985) and Draper, et al., *Plant Cell Physiol.*, 23:451 (1982). Electroporation of protoplasts and whole cells and tissues has also been described (D'Halluin, et al., *Plant Cell*, 4:1495-1505 (1992); and Spencer, et al., *Plant Mol. Biol.*, 24:51-61 (1994)).

Following transformation of potato target tissues, expression of the above-described selectable marker genes allows for preferential selection of transformed cells, tissues, and/or plants, using regeneration and selection methods well known in the art.

The foregoing methods for transformation may be used for producing a transgenic variety. The transgenic variety could then be crossed with another (non-transformed or

transformed) variety in order to produce a new transgenic variety. Alternatively, a genetic trait that has been engineered into a particular potato line using the foregoing transformation techniques could be moved into another line using traditional backcrossing techniques that are well known in the plant breeding arts. For example, a backcrossing approach could be used to move an engineered trait from a public, non-elite variety into an elite variety, or from a variety containing a foreign gene in its genome into a variety or varieties that do not contain that gene. As used herein, "crossing" can refer to a simple x by y cross or the process of backcrossing depending on the context.

Likewise, by means of one embodiment, agronomic genes can be expressed in transformed plants. More particularly, plants can be genetically engineered to express various phenotypes of agronomic interest. Exemplary genes implicated in this regard include, but are not limited to, those categorized below:

1. Genes that Confer Resistance to Pests or Disease and that Encode:

A. Plant disease resistance genes. Plant defenses are often activated by specific interaction between the product of a disease resistance gene (R) in the plant and the product of a corresponding avirulence (Avr) gene in the pathogen. A plant variety can be transformed with one or more cloned resistance genes to engineer plants that are resistant to specific pathogen strains. See, for example, Jones, et al., *Science*, 266:789 (1994) (cloning of the tomato Cf-9 gene for resistance to *Cladosporium fulvum*); Martin, et al., *Science*, 262:1432 (1993) (tomato Pto gene for resistance to *Pseudomonas syringae* pv. tomato encodes a protein kinase); Mindrinos, et al., *Cell*, 78:1089 (1994) (*Arabidopsis* RSP2 gene for resistance to *Pseudomonas syringae*); McDowell & Woffenden, *Trends Biotechnol.*, 21(4): 178-83 (2003); and Toyoda, et al., *Transgenic Res.*, 11 (6):567-82 (2002).

B. A gene conferring resistance to a pest, such as the Colorado potato beetle. See, for example, Mi, X., et al., "Transgenic potato plants expressing cry3A gene confer resistance to Colorado potato beetle" *C. R. Biol.* 2015 July, 338(7):443-50; and the potato tuber moth, Davidson, M. M., et al., "Development and Evaluation of Potatoes Transgenic for a cryAc1 Gene Conferring Resistance to Potato Tuber Moth" *J. Amer. Soc. Hort. Sci.* 127(4):590-596.

C. A *Bacillus thuringiensis* protein, a derivative thereof or a synthetic polypeptide modeled thereon. See, for example, Geiser, et al., *Gene*, 48:109 (1986), who disclose the cloning and nucleotide sequence of a Bt δ -endotoxin gene. Moreover, DNA molecules encoding δ -endotoxin genes can be purchased from American Type Culture Collection, Manassas, Va., for example, under ATCC Accession Nos. 40098, 67136, 31995, and 31998.

D. A lectin. See, for example, Van Damme, et al., *Plant Molec. Biol.*, 24:25 (1994), who disclose the nucleotide sequences of several *Clivia miniata* mannose-binding lectin genes.

E. A vitamin-binding protein such as avidin. See, International Application No. PCT/US1993/006487, which teaches the use of avidin and avidin homologues as larvicides against insect pests.

F. An enzyme inhibitor, for example, a protease or proteinase inhibitor or an amylase inhibitor. See, for example, Abe, et al., *J. Biol. Chem.*, 262: 16793 (1987) (nucleotide sequence of rice cysteine proteinase inhibitor); Huub, et al., *Plant Molec. Biol.*, 21:985 (1993) (nucleotide sequence of cDNA encoding tobacco proteinase inhibitor I); Sumitani, et al., *Biosci. Biotech. Biochem.*, 57:1243 (1993) (nucleotide

sequence of *Streptomyces nitrosporeus* α -amylase inhibitor); and U.S. Pat. No. 5,494,813.

G. An insect-specific hormone or pheromone, such as an ecdysteroid or juvenile hormone, a variant thereof, a mimetic based thereon, or an antagonist or agonist thereof. See, for example, the disclosure by Hammock, et al., *Nature*, 344:458 (1990), of baculovirus expression of cloned juvenile hormone esterase, an inactivator of juvenile hormone.

H. An insect-specific peptide or neuropeptide which, upon expression, disrupts the physiology of the affected pest. For example, see the disclosures of Regan, *J. Biol. Chem.*, 269:9 (1994) (expression cloning yields DNA coding for insect diuretic hormone receptor); Pratt, et al., *Biochem. Biophys. Res. Comm.*, 163:1243 (1989) (an allostatin is identified in *Diploptera puntata*); Chattopadhyay, et al., *Critical Reviews in Microbiology*, 30(1):33-54 (2004); Zjawiony, *J. Nat. Prod.*, 67(2):300-310 (2004); Carlini & Grossi-de-Sa, *Toxicol.*, 40(11):1515-1539 (2002); Ussuf, et al., *Curr Sci.*, 80(7):847-853 (2001); Vasconcelos & Oliveira, *Toxicol.*, 44(4):385-403 (2004). See also, U.S. Pat. No. 5,266,317 which discloses genes encoding insect-specific, paralytic neurotoxins.

I. An insect-specific venom produced in nature by a snake, a wasp, etc. For example, see, Pang, et al., *Gene*, 116:165 (1992), for disclosure of heterologous expression in plants of a gene coding for a scorpion insectotoxic peptide.

J. An enzyme responsible for a hyperaccumulation of a monoterpene, a sesquiterpene, a steroid, hydroxamic acid, a phenylpropanoid derivative, or another non-protein molecule with insecticidal activity.

K. An enzyme involved in the modification, including the post-translational modification, of a biologically active molecule; for example, a glycolytic enzyme, a proteolytic enzyme, a lipolytic enzyme, a nuclease, a cyclase, a transaminase, an esterase, a hydrolase, a phosphatase, a kinase, a phosphorylase, a polymerase, an elastase, a chitinase, and a glucanase, whether natural or synthetic. See, U.S. Pat. No. 5,955,653 which discloses the nucleotide sequence of a callase gene. DNA molecules which contain chitinase-encoding sequences can be obtained, for example, from the ATCC under Accession Nos. 39637 and 67152. See also, Kramer, et al., *Insect Biochem. Molec. Biol.*, 23:691 (1993), who teach the nucleotide sequence of a cDNA encoding tobacco hornworm chitinase, and Kawalleck, et al., *Plant Molec. Biol.*, 21:673 (1993), who provide the nucleotide sequence of the parsley ubi4-2 polyubiquitin gene, U.S. Pat. Nos. 7,145,060, 7,087,810, and 6,563,020.

L. A hydrophobic moment peptide. See, U.S. Pat. No. 5,580,852, which discloses peptide derivatives of tachyplesin which inhibit fungal plant pathogens, and U.S. Pat. No. 5,607,914 which teaches synthetic antimicrobial peptides that confer disease resistance.

M. A membrane permease, a channel former or a channel blocker. For example, see the disclosure of Jaynes, et al., *Plant Sci.*, 89:43 (1993), of heterologous expression of a cecropin-13 lytic peptide analog to render transgenic tobacco plants resistant to *Pseudomonas solanacearum*.

N. A viral-invasive protein or a complex toxin derived therefrom. For example, the accumulation of viral coat proteins in transformed plant cells imparts resistance to viral infection and/or disease development effected by the virus from which the coat protein gene is derived, as well as by related viruses. See, Beachy, et al., *Ann. Rev. Phytopathol.*, 28:451 (1990). Coat protein-mediated resistance has been conferred upon transformed plants against alfalfa mosaic virus, cucumber mosaic virus, and tobacco mosaic virus.

O. An insect-specific antibody or an immunotoxin derived therefrom. Thus, an antibody targeted to a critical metabolic function in the insect gut would inactivate an affected enzyme, killing the insect.

P. A virus-specific antibody. See, for example, Tavladoraki, et al., *Nature*, 366:469 (1993), who show that transgenic plants expressing recombinant antibody genes are protected from virus attack.

Q. A developmental-arrestive protein produced in nature by a pathogen or a parasite. Thus, fungal endo- α -1,4-D-polygalacturonases facilitate fungal colonization and plant nutrient release by solubilizing plant cell wall homo- α -1,4-D-galacturonase. See, Lamb, et al., *Bio Technology*, 10:1436 (1992). The cloning and characterization of a gene which encodes a bean endopolygalacturonase-inhibiting protein is described by Toubart, et al., *Plant J.*, 2:367 (1992).

R. A developmental-arrestive protein produced in nature by a plant. For example, Logemann, et al., *Bio Technology*, 10:305 (1992), have shown that transgenic plants expressing the barley ribosome-inactivating gene have an increased resistance to fungal disease.

S. Genes involved in the Systemic Acquired Resistance (SAR) Response and/or the pathogenesis-related genes. Briggs, S., *Current Biology*, 5(2) (1995); Pieterse & Van Loon, *Curr. Opin. Plant Bio.*, 7(4):456-64 (2004); and Somssich, *Cell*, 113(7):815-6 (2003).

T. Antifungal genes. See, Cornelissen and Melchers, *Plant Physiol.*, 101:709-712 (1993); Parijs, et al., *Planta*, 183:258-264 (1991); and Bushnell, et al., *Can. J of Plant Path.*, 20(2): 137-149 (1998). See also, U.S. Pat. No. 6,875,907.

U. Detoxification genes, such as for fumonisin, beauvericin, moniliformin, and zearalenone and their structurally-related derivatives. See, U.S. Pat. No. 5,792,931.

V. Cystatin and cysteine proteinase inhibitors. See, U.S. Pat. No. 7,205,453.

W. Defensin genes. See, U.S. Pat. Nos. 6,911,577, 7,855,327, 7,855,328, 7,897,847, 7,910,806, 7,919,686, and 8,026,415.

X. Genes conferring resistance to nematodes. See, U.S. Pat. Nos. 5,994,627 and 6,294,712; Urwin, et al., *Planta*, 204:472-479 (1998); Williamson, *Curr Opin Plant Bio.*, 2(4): 327-31 (1999).

Y. Genes conferring resistance to potato late blight, such as Rpi-Vnt1, which is well-known in art.

Z. Genes conferring resistance to potato leaf roll virus (PLRV) through gene silencing mechanism, such as plrv orf1 and 2, which is well-known in art.

AA. Genes conferring resistance to potato virus Y (PVY) through "pathogen-derived resistance" mechanism, such as pvy cp, which is well-known in art. Please see Song, Ye-Su, "Genetic marker analysis in potato for extreme resistance (Rysto) to PVY and for chip quality after long term storage at 4° C." Dissertation, Technical University of Munchen, dated Jul. 26, 2004.

Any of the above-listed disease or pest resistance genes (A-AA) can be introduced into the claimed potato cultivar through a variety of means including, but not limited to, transformation and crossing.

2. Genes that Confer Resistance to an Herbicide, for Example:

A. An herbicide that inhibits the growing point or meristem, such as an imidazolinone or a sulfonylurea. Exemplary genes in this category code for mutant ALS and AHAS enzyme as described, for example, by Lee, et al., *EMBO J.*, 7:1241 (1988) and Mild, et al., *Theor. Appl. Genet.*, 80:449 (1990), respectively.

B. Glyphosate (resistance conferred by mutant 5-enolpyruvylshikimate-3-phosphate synthase (EPSPS) and aroA genes, respectively) and other phosphono compounds, such as glufosinate (phosphinothricin acetyl transferase (PAT) and *Streptomyces hygroscopicus* PAT bar genes), pyridinoxy or phenoxy propionic acids, and cyclohexanediones (ACCase inhibitor-encoding genes). See, for example, U.S. Pat. No. 4,940,835 which discloses the nucleotide sequence of a form of EPSPS which can confer glyphosate resistance. U.S. Pat. No. 5,627,061 which describes genes encoding EPSPS enzymes. See also, U.S. Pat. Nos. 6,566,587, 6,338,961, 6,248,876, 6,040,497, 5,804,425, 5,633,435, 5,145,783, 4,971,908, 5,312,910, 5,188,642, 4,940,835, 5,866,775, 6,225,114, 6,130,366, 5,310,667, 4,535,060, 4,769,061, 5,633,448, 5,510,471, 6,803,501, RE 36,449, RE 37,287, and 5,491,288, which are incorporated herein by reference for this purpose. Glyphosate resistance is also imparted to plants that express a gene that encodes a glyphosate oxido-reductase enzyme, as described more fully in U.S. Pat. Nos. 5,776,760 and 5,463,175, which are incorporated herein by reference for this purpose. In addition, glyphosate resistance can be imparted to plants by the over expression of genes encoding glyphosate N-acetyltransferase. See, for example, U.S. Pat. No. 7,462,481. A DNA molecule encoding a mutant aroA gene can be obtained under ATCC Accession No. 39256, and the nucleotide sequence of the mutant gene is disclosed in U.S. Pat. No. 4,769,061. European Patent Appl. No. 0333033 and U.S. Pat. No. 4,975,374 disclose nucleotide sequences of glutamine synthetase genes which confer resistance to herbicides such as L-phosphinothricin. The nucleotide sequence of a PAT gene is provided in European Patent No. 0242246 to Leemans, et al. DeGreef, et al., *Bio Technology*, 7:61 (1989) describe the production of transgenic plants that express chimeric bar genes coding for phosphinothricin acetyl transferase activity. Exemplary of genes conferring resistance to phenoxy propionic acids and cyclohexones, such as sethoxydim and haloxyfop are the Acc1-S1, Acc1-S2, and Acc2-S3 genes described by Marshall, et al., *Theor. Appl. Genet.*, 83:435 (1992).

C. An herbicide that inhibits photosynthesis, such as a triazine (psbA and gs+ genes) and a benzonitrile (nitrilase gene). Przibila, et al., *Plant Cell*, 3:169 (1991), describe the transformation of *Chlamydomonas* with plasmids encoding mutant psbA genes. Nucleotide sequences for nitrilase genes are disclosed in U.S. Pat. No. 4,810,648 to Stalker and DNA molecules containing these genes are available under ATCC Accession Nos. 53435, 67441, and 67442. Cloning and expression of DNA coding for a glutathione S-transferase is described by Hayes, et al., *Biochem. J.*, 285:173 (1992). Protoporphyrinogen oxidase (PPO) is the target of the PPO-inhibitor class of herbicides; a PPO-inhibitor resistant PPO gene was recently identified in *Amaranthus tuberculatus* (Patzoldt et al., *PNAS*, 103(33):12329-2334, 2006). The herbicide methyl viologen inhibits CO₂ assimilation. Foyer et al. (*Plant Physiol.*, 109:1047-1057, 1995) describe a plant overexpressing glutathione reductase (GR) which is resistant to methyl viologen treatment. Bromoxynil resistance by introducing a chimeric gene containing the bxn gene (*Science*, 242(4877): 419-23, 1988).

D. Acetohydroxy acid synthase, which has been found to make plants that express this enzyme resistant to multiple types of herbicides, has been introduced into a variety of plants. See, Hattori, et al., *Mol. Gen. Genet.*, 246:419 (1995). Other genes that confer tolerance to herbicides include a gene encoding a chimeric protein of rat cytochrome P4507A1 and yeast NADPH-cytochrome P450

oxidoreductase (Shiota, et al., *Plant Physiol.*, 106:17 (1994)); genes for glutathione reductase and superoxide dismutase (Aono, et al., *Plant Cell Physiol.*, 36:1687 (1995)); and genes for various phosphotransferases (Datta, et al., *Plant Mol. Biol.*, 20:619 (1992)).

E. Protoporphyrinogen oxidase (protox) is necessary for the production of chlorophyll, which is necessary for all plant survival. The protox enzyme serves as the target for a variety of herbicidal compounds. These herbicides also inhibit growth of all the different species of plants present, causing their total destruction. The development of plants containing altered protox activity which are resistant to these herbicides are described in U.S. Pat. Nos. 6,288,306, 6,282,837, 5,767,373, and 6,084,155.

Any of the above listed herbicide genes (A-E) can be introduced into the claimed potato cultivar through a variety of means including but not limited to transformation and crossing.

3. Genes that Confer or Contribute to a Value-Added Trait, Such as:

A. Modified fatty acid metabolism, for example, by transforming a plant with an antisense gene of stearyl-ACP desaturase to increase stearic acid content of the plant. See, Knultzon, et al., *Proc. Natl. Acad. Sci. USA*, 89:2625 (1992).

B. Decreased phytate content: 1) Introduction of a phytase-encoding gene enhances breakdown of phytate, adding more free phosphate to the transformed plant. For example, see, Van Hartingsveldt, et al., *Gene*, 127:87 (1993), for a disclosure of the nucleotide sequence of an *Aspergillus niger* phytase gene. 2) Up-regulation of a gene that reduces phytate content. In maize, this, for example, could be accomplished by cloning and then re-introducing DNA associated with one or more of the alleles, such as the LPA alleles, identified in maize mutants characterized by low levels of phytic acid, such as in Raboy, et al., *Maydica*, 35:383 (1990), and/or by altering inositol kinase activity as in, for example, U.S. Pat. Nos. 7,425,442, 7,714,187, 6,197,561, 6,2191,224, 6,855,869, 6,391,348, 6,197,561, and 6,291,224; U.S. Publ. Nos. 2003/000901, 2003/0009011, and 2006/272046; and International Pub. Nos. WO 98/45448, and WO 01/04147.

C. Modified carbohydrate composition effected, for example, by transforming plants with a gene coding for an enzyme that alters the branching pattern of starch, or a gene altering thioredoxin, such as NTR and/or TRX (See, U.S. Pat. No. 6,531,648, which is incorporated by reference for this purpose), and/or a gamma zein knock out or mutant, such as cs27 or TUSC27 or en27 (See, U.S. Pat. Nos. 6,858,778, 7,741,533 and U.S. Publ. No. 2005/0160488, which are incorporated by reference for this purpose). See, Shiroza, et al., *J. Bacteriol.*, 170:810 (1988) (nucleotide sequence of *Streptococcus mutans* fructosyltransferase gene); Steinmetz, et al., *Mol. Gen. Genet.*, 200:220 (1985) (nucleotide sequence of *Bacillus subtilis* levansucrase gene); Pen, et al., *Bio Technology*, 10:292 (1992) (production of transgenic plants that express *Bacillus licheniformis* α -amylase); Elliot, et al., *Plant Molec. Biol.*, 21:515 (1993) (nucleotide sequences of tomato invertase genes); Sogaard, et al., *J. Biol. Chem.*, 268:22480-22484 (1993) (site-directed mutagenesis of barley α -amylase gene); Fisher, et al., *Plant Physiol.*, 102:1045 (1993) (maize endosperm starch branching enzyme II); International Pub. No. WO 99/10498 (improved digestibility and/or starch extraction through modification of UDP-D-xylose 4-epimerase, Fragile 1 and 2, Ref 1, HCHL, C4H); U.S. Pat. No. 6,232,529 (method of producing high oil seed by modification of starch levels (AGP)). The fatty acid modification genes mentioned above may also

be used to affect starch content and/or composition through the interrelationship of the starch and oil pathways.

D. Elevated oleic acid via FAD-2 gene modification and/or decreased linolenic acid via FAD-3 gene modification. See, U.S. Pat. Nos. 5,952,544, 6,063,947, and 6,323,392.

E. Altering conjugated linolenic or linoleic acid content, such as in U.S. Pat. No. 6,593,514. Altering LEC1, AGP, Dek1, Superal1, milps, and various lpa genes, such as lpa1, lpa3, lpt, or hgg. See, for example, U.S. Pat. Nos. 7,122,658, 7,342,418, 6,232,529, 7,888,560, 6,423,886, 6,197,561, 6,825,397 and 7,157,621; U.S. Publ. No. 2003/0079247; International Publ. No. WO 2003/011015; and Rivera-Madrid, R., et al., *Proc. Natl. Acad. Sci.*, 92:5620-5624 (1995).

F. Altered antioxidant content or composition, such as alteration of tocopherol or tocotrienols. See, for example, U.S. Pat. Nos. 6,787,683, 7,154,029 and International Publ. No. WO 00/68393 (involving the manipulation of antioxidant levels through alteration of a phytyl prenyl transferase (ppt)); and U.S. Pat. Nos. 7,154,029 and 7,622,658 (through alteration of a homogentisate geranyl geranyl transferase (hgg)).

G. Altered essential seed amino acids. See, for example, U.S. Pat. No. 6,127,600 (method of increasing accumulation of essential amino acids in seeds); U.S. Pat. No. 6,080,913 (binary methods of increasing accumulation of essential amino acids in seeds); U.S. Pat. No. 5,990,389 and International Publ. No. WO 95/15392 (high lysine); U.S. Pat. No. 5,850,016 (alteration of amino acid compositions in seeds); U.S. Pat. No. 5,885,802 (high methionine); U.S. Pat. No. 5,885,801 and International Publ. No. WO96/01905 (high threonine); U.S. Pat. Nos. 6,664,445, 7,022,895, 7,368,633, and 7,439,420 (plant amino acid biosynthetic enzymes); U.S. Pat. No. 6,459,019 and U.S. application Ser. No. 09/381,485 (increased lysine and threonine); U.S. Pat. No. 6,441,274 (plant tryptophan synthase beta subunit); U.S. Pat. No. 6,346,403 (methionine metabolic enzymes); U.S. Pat. No. 5,939,599 (high sulfur); U.S. Pat. No. 5,912,414 (increased methionine); U.S. Pat. No. 5,633,436 (increasing sulfur amino acid content); U.S. Pat. No. 5,559,223 (synthetic storage proteins with defined structure containing programmable levels of essential amino acids for improvement of the nutritional value of plants); U.S. Pat. No. 6,194,638 (hemicellulose); U.S. Pat. No. 7,098,381 (UDPGdH); U.S. Pat. No. 6,194,638 (RGP); U.S. Pat. Nos. 6,399,859, 6,930,225, 7,179,955, 6,803,498, 5,850,016, and 7,053,282 (alteration of amino acid compositions in seeds); WO 99/29882 (methods for altering amino acid content of proteins); U.S. application Ser. No. 09/297,418 (proteins with enhanced levels of essential amino acids); WO 98/45458 (engineered seed protein having higher percentage of essential amino acids); WO 01/79516; and U.S. Pat. Nos. 6,803,498, 6,930,225, 7,307,149, 7,524,933, 7,579,443, 7,838,632, 7,851,597, and 7,982,009 (maize cellulose synthases).

4. Genes that Control Male Sterility:

There are several methods of conferring genetic male sterility in potatoes. For example, male sterility occurs more often in tetraploid cultivars and related taxa. Please see Grun P., et al., "Multiple differentiation of plasmons of diploid species of *Solanum*." *Genetics* 47: 1321-1333 (1962). The male sterility is a consequence of nuclear-cytoplasm interactions, where the dominant Ms gene interacts with the cytoplasm of *S. tuberosum* to cause male sterility and the dominant Ri gene restores fertility. Please see Iwanaga M., et al., "A restorer gene for genetic-cytoplasmic male sterility in cultivated potatoes". *Am. Potato J.* 68: 19-28 (1991a).

5. Genes that Create a Site for Site Specific DNA Integration:

This includes the introduction of FRT sites that may be used in the FLP/FRT system and/or Lox sites that may be used in the Cre/Loxp system. See, for example, Lyznik, et al., Site-Specific Recombination for Genetic Engineering in Plants, *Plant Cell Rep.*, 21:925-932 (2003) and U.S. Pat. No. 6,187,994, which are hereby incorporated by reference. Other systems that may be used include the Gin recombinase of phage Mu (Maeser, et al. (1991); Vicki Chandler, The Maize Handbook, Ch. 118 (Springer-Verlag 1994)); the Pin recombinase of *E. coli* (Enomoto, et al. (1983)); and the R/Rs system of the pSRi plasmid (Araki, et al. (1992)).

6. Genes that Affect Abiotic Stress Resistance:

Genes that affect abiotic stress resistance (including but not limited to flowering, enhancement of nitrogen utilization efficiency, altered nitrogen responsiveness, drought resistance or tolerance, cold resistance or tolerance, and salt resistance or tolerance) and increased yield under stress. For example, see U.S. Pat. No. 6,653,535 where water use efficiency is altered through alteration of malate; U.S. Pat. Nos. 5,892,009, 5,965,705, 5,929,305, 5,891,859, 6,417,428, 6,664,446, 6,706,866, 6,717,034, 6,801,104, 6,946,586, 7,238,860, 7,635,800, 7,135,616, 7,193,129, and 7,601,893; and International Publ. Nos. WO 2001/026459, WO 2001/035725, WO 2001/035727, WO 2001/036444, WO 2001/036597, WO 2001/036598, WO 2002/015675, and WO 2002/077185, describing genes, including CBF genes and transcription factors effective in mitigating the negative effects of freezing, high salinity, and drought on plants, as well as conferring other positive effects on plant phenotype; U.S. Publ. No. 2004/0148654, where abscisic acid is altered in plants resulting in improved plant phenotype, such as increased yield and/or increased tolerance to abiotic stress; U.S. Pat. Nos. 6,992,237, 6,429,003, 7,049,115, and 7,262,038, where cytokinin expression is modified resulting in plants with increased stress tolerance, such as drought tolerance, and/or increased yield. See also, WO 02/02776, WO 2003/052063, JP 2002281975, U.S. Pat. No. 6,084,153, WO 01/64898, and U.S. Pat. Nos. 6,177,275 and 6,107,547 (enhancement of nitrogen utilization and altered nitrogen responsiveness). For ethylene alteration, see, U.S. Publ. Nos. 2004/0128719, 2003/0166197, and U.S. application Ser. No. 09/856,834. For plant transcription factors or transcriptional regulators of abiotic stress, see, e.g., U.S. Publ. Nos. 2004/0098764 or 2004/0078852.

Other genes and transcription factors that affect plant growth and agronomic traits, such as yield, flowering, plant growth, and/or plant structure, can be introduced or introgressed into plants. See for example, U.S. Pat. Nos. 6,140,085, and 6,265,637 (CO); U.S. Pat. No. 6,670,526 (ESD4); U.S. Pat. Nos. 6,573,430 and 7,157,279 (TFL); U.S. Pat. No. 6,713,663 (FT); U.S. Pat. Nos. 6,794,560, 6,307,126 (GAI); U.S. Pat. No. 7,045,682 (VRN1); U.S. Pat. Nos. 6,949,694 and 7,253,274 (VRN2); U.S. Pat. No. 6,887,708 (GI); U.S. Pat. No. 7,320,158 (FRI); U.S. Pat. No. 6,307,126 (GAI); U.S. Pat. Nos. 6,762,348 and 7,268,272 (D8 and Rht); and U.S. Pat. Nos. 7,345,217, 7,511,190, 7,659,446, and 7,825,296 (transcription factors).

Tissue Culture

Further reproduction of the variety can occur by tissue culture and regeneration. Tissue culture of various tissues of potatoes and regeneration of plants therefrom is well-known and widely published. See, Ahloowalia, B. S., "Plant regeneration from callus culture in potato" *Euphytica*. 31(3): pp 755-759 (December 1982); and Wang, P. J., "Regeneration of Virus-free Potato from Tissue Culture" in *Plant Tissue*

Culture and Its Bio-technological Application, Bartz, et al. (eds). Springer-Verlag Berlin Heidelberg. pp 386-391 (1977). Thus, another aspect or embodiment is to provide cells which upon growth and differentiation produce potato plants having the physiological and morphological characteristics of potato cultivar FL 2505.

Regeneration refers to the development of a plant from tissue culture. The term "tissue culture" indicates a composition comprising isolated cells of the same or a different type or a collection of such cells organized into parts of a plant. Exemplary types of tissue cultures are protoplasts, calli, plant clumps, and plant cells that can generate tissue culture that are intact in plants or parts of plants, such as embryos, pollen, flowers, seeds, pods, petioles, leaves, stems, roots, root tips, anthers, pistils, and the like. Means for preparing and maintaining plant tissue culture are well known in the art. By way of example, a tissue culture comprising organs has been used to produce regenerated plants. U.S. Pat. Nos. 5,959,185, 5,973,234, and 5,977,445 describe certain techniques, the disclosures of which are incorporated herein by reference.

INDUSTRIAL USES

Potato has a wide variety of uses in the commodity area. For example, fresh potatoes can be cooked (fried, baked, boiled, etc). Potatoes can be used to make potato chips, frozen potato items such as hash/home fries/French fries, dehydrated potato flakes, potato granules, ingredients in food snacks, potato flour, potato starch, and alcoholic beverages, as well as non-food uses such as potato starch used by the pharmaceutical, textile, wood, and paper industries as an adhesive, binder, texture agent, and filler, and by oil drilling firms to wash boreholes. Potato starch can also be used in place of polystyrene and other plastics disposable dishes and utensils. Potato peel and other wastes from potato processing can be liquefied and fermented to produce fuel-grade ethanol. Thus, a further embodiment provides for a food product or non-food product made from a part of the potato plant variety FL 2505. The food product may be a French fry, potato chip, dehydrated potato material, potato flakes, or potato granules.

In addition to the exemplary aspects and embodiments described above, further aspects and embodiments will become apparent by study of the following descriptions.

While a number of exemplary aspects and embodiments have been discussed above, those of skill in the art will recognize certain modifications, permutations, additions and sub-combinations thereof. It is therefore intended that the following appended claims and claims hereafter introduced are interpreted to include all such modifications, permutations, additions, and sub-combinations as are within their true spirit and scope.

One embodiment may be embodied in other specific forms without departing from its spirit or essential characteristics. The described embodiments are to be considered in all respects only as illustrative and not restrictive. All changes which come within the meaning and range of equivalency of the claims are to be embraced within their scope.

Various embodiments, include components, methods, processes, systems and/or apparatus substantially as depicted and described herein, including various embodiments, sub-combinations, and subsets thereof. Those of skill in the art will understand how to make and use an embodiment(s) after understanding the present disclosure.

The foregoing discussion of the embodiments has been presented for purposes of illustration and description. The foregoing is not intended to limit the embodiments to the form or forms disclosed herein. In the foregoing Detailed Description for example, various features of the embodiments are grouped together in one or more embodiments for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the embodiment(s) requires more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive aspects lie in less than all features of a single foregoing disclosed embodiment. Thus, the following claims are hereby incorporated into this Detailed Description.

Moreover, though the description of the embodiments has included description of one or more embodiments and certain variations and modifications, other variations and modifications are within the scope of the embodiments (e.g., as may be within the skill and knowledge of those in the art, after understanding the present disclosure). It is intended to obtain rights which include alternative embodiments to the extent permitted, including alternate, interchangeable and/or equivalent structures, functions, ranges or acts to those claimed, whether or not such alternate, interchangeable and/or equivalent structures, functions, ranges or acts are disclosed herein, and without intending to publicly dedicate any patentable subject matter.

The terms “comprising,” “having,” “including,” and “containing” are to be construed as open-ended terms (i.e., meaning “including, but not limited to,”) unless otherwise noted. Recitation of ranges of values herein are merely intended to serve as a shorthand method of referring individually to each separate value falling within the range, unless otherwise indicated herein, and each separate value is incorporated into the specification as if it were individually recited herein. For example, if the range 10-15 is disclosed, then 11, 12, 13, and 14 are also disclosed. All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (e.g., “such as”) provided herein, is intended merely to better illuminate the embodiments and does not pose a limitation on the scope of the embodiments unless otherwise claimed.

DEPOSIT INFORMATION

A microtuber deposit of the Frito-Lay North America, Inc. proprietary potato variety FL 2505 disclosed above and recited in the appended claims is maintained by Frito-Lay North America, Inc., 7701 Legacy Drive, Plano, Texas 75024. A deposit has been made with the Provasoli-Guillard National Center for Marine Algae and Microbiota (NCMA), 60 Bigelow

Drive, East Boothbay, ME 04544, USA, with NCMA Accession No. 202404002 on Apr. 2, 2024. Access to this deposit will be available during the pendency of this application to persons determined by the Commissioner of Patents and Trademarks to be entitled thereto under 37 C.F.R. 1.14 and 35 U.S.C. § 122. Upon allowance of any claims in this application, all restrictions on the availability to the public of the variety will be irrevocably removed by affording access to a deposit of the same variety with a Budapest Treaty depository. The deposit will be maintained in the depository for a period of 30 years, or 5 years after the last request, or for the effective life of the patent, whichever is longer, and will be replaced if necessary during that period.

Applicant has no authority to waive any restrictions imposed by law on the transfer of biological material or its transportation in commerce. Applicant does not waive any infringement of rights granted under this patent or under the Plant Variety Protection Act (7 USC 2321 et seq.). Unauthorized seed multiplication is prohibited.

What is claimed is:

1. A plant of potato cultivar FL 2505, wherein a representative sample of plant tissue of said variety was deposited under NCMA No. 202404002.

2. A potato plant, or a part thereof, produced by growing the plant of claim 1, wherein the plant part comprises at least one cell of potato cultivar FL 2505.

3. A potato plant having all of the physiological and morphological characteristics of the plant of claim 2.

4. A tissue culture of cells produced from the plant of claim 2, wherein said cells of the tissue culture are produced from a plant part selected from the group consisting of tuber, leaf, pollen, embryo, cotyledon, hypocotyl, meristematic cell, root, root tip, pistil, anther, flowers, ovule, light sprout, petiole, eye, stem, and tuber.

5. A potato plant regenerated from the tissue culture of claim 4, wherein said plant has all of the physiological and morphological characteristics of potato cultivar FL 2505.

6. A potato light sprout produced by growing the plant of claim 1.

7. A potato plant, or a part thereof, produced by growing the light sprout of claim 6.

8. A potato plant regenerated from tissue culture of the potato plant of claim 7, wherein said potato plant has all of the physiological and morphological characteristics of potato cultivar FL 2505.

9. A method for producing an F₁ progeny potato plant, said method comprising crossing two potato plants and harvesting the resultant potato seed, wherein at least one potato plant is the potato plant of claim 1.

10. A method of introducing a desired trait into potato cultivar FL 2505, wherein the method comprises:

(a) crossing a FL 2505 plant, wherein a representative sample of tubers was deposited under NCMA No. 202404002, with a plant of another potato cultivar that comprises a desired trait to produce progeny plants, wherein the desired trait is selected from the group consisting of herbicide resistance, insect resistance, resistance for bacterial, fungal, or viral disease, male fertility, male sterility, enhanced nutritional quality, modified fatty acid metabolism, modified carbohydrate metabolism, modified yield, modified glycoalkaloid content, and industrial usage;

(b) selecting one or more progeny plants that have the desired trait using one or more markers for the desired trait;

(c) backcrossing the selected progeny plants with FL 2505 plants to produce backcross progeny plants;

(d) selecting for backcross progeny plants that have the desired trait and one or more markers for the desired trait; and

(e) repeating steps (c) and (d) two or more times in succession to produce selected third or higher backcross progeny plants that comprise the desired trait and one or more markers.

11. A potato plant produced by the method of claim 10.

12. The potato plant of claim 11, wherein the desired trait is herbicide resistance and the resistance is conferred to an herbicide selected from the group consisting of imidazolinone, sulfonylurea, glyphosate, glufosinate, L-phosphinothricin, triazine and benzonitrile.

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13. The potato plant of claim 11, wherein the desired trait is insect resistance and the insect resistance is conferred by a transgene encoding a *Bacillus thuringiensis* endotoxin.

14. The potato plant of claim 11, wherein the desired trait is modified fatty acid metabolism or modified carbohydrate metabolism and said desired trait is conferred by a nucleic acid encoding a protein selected from the group consisting of fructosyltransferase, levansucrase, alpha-amylase, invertase, and starch branching enzyme or DNA encoding an antisense of stearyl-ACP desaturase.

15. A method of producing a commodity plant product, comprising obtaining the plant of claim 1, or a part thereof, and producing the commodity plant product from said plant or plant part thereof, wherein said commodity plant product is selected from the group consisting of french fries, potato chips, dehydrated potato material, potato flakes and potato granules.

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16. The commodity plant product produced by the method of claim 15.

17. A method of vegetatively propagating the plant of claim 1, comprising the steps of:

collecting tissue or cells capable of being propagated from said plant;

cultivating said tissue or cells to obtain proliferated shoots; and

rooting said proliferated shoots to obtain rooted plantlets; or cultivating said tissue or cells to obtain proliferated shoots, or to obtain plantlets.

18. A potato plant produced by growing the plantlets or proliferated shoots of claim 17.

19. A method for producing an embryo or seed, wherein the method comprises crossing the plant of claim 1 with another plant and harvesting the resultant embryo or seed.

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